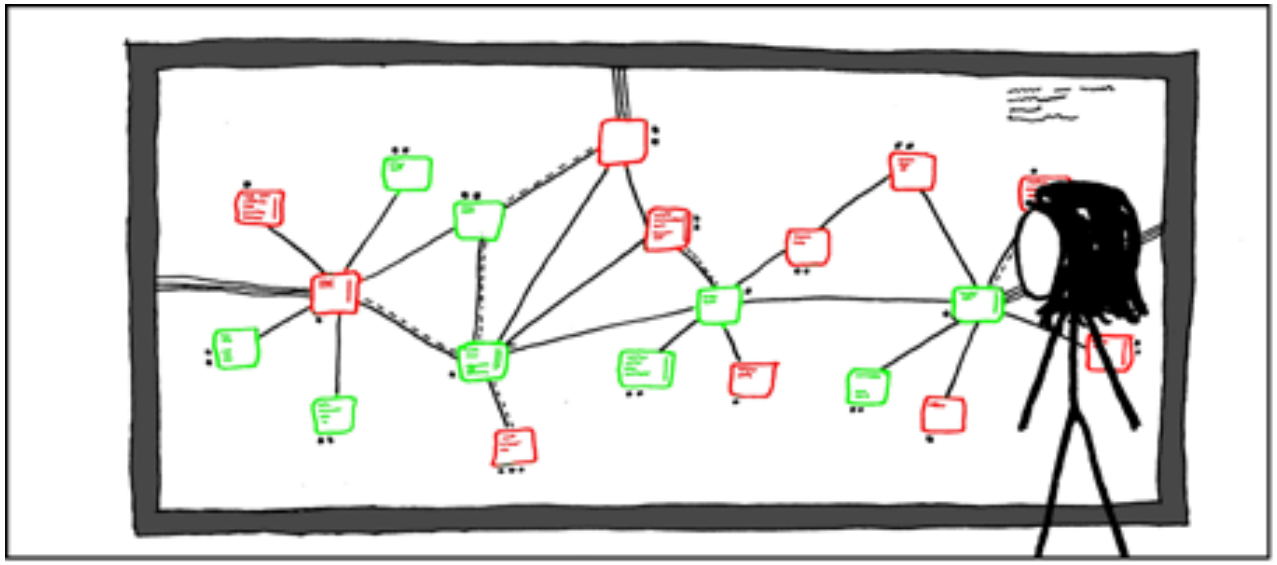
Virtual networking East/West Performance

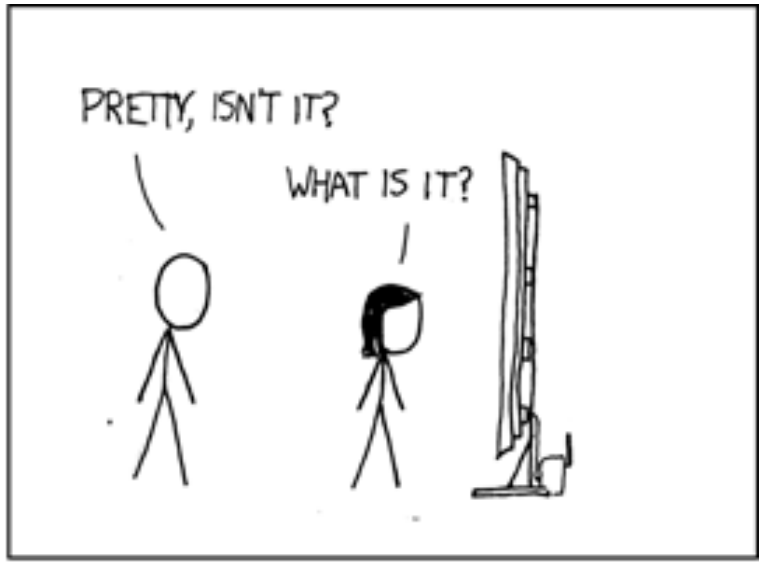
Do Offloads really matter?

Stephen Hemminger
stephen@networkplumber.org

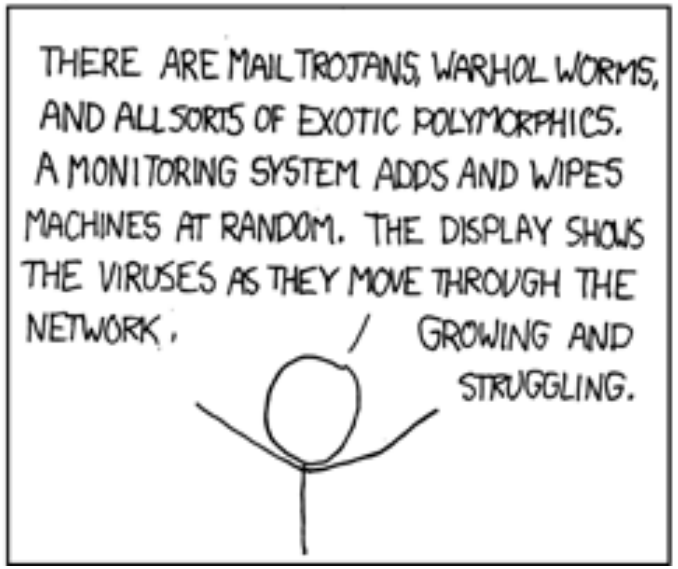
Agenda


Why?
Current state
Use Cases
What?
Tests
Tools
Techniques
Results
Measurements
Good and Bad
Future
Next steps
Crazy ideas



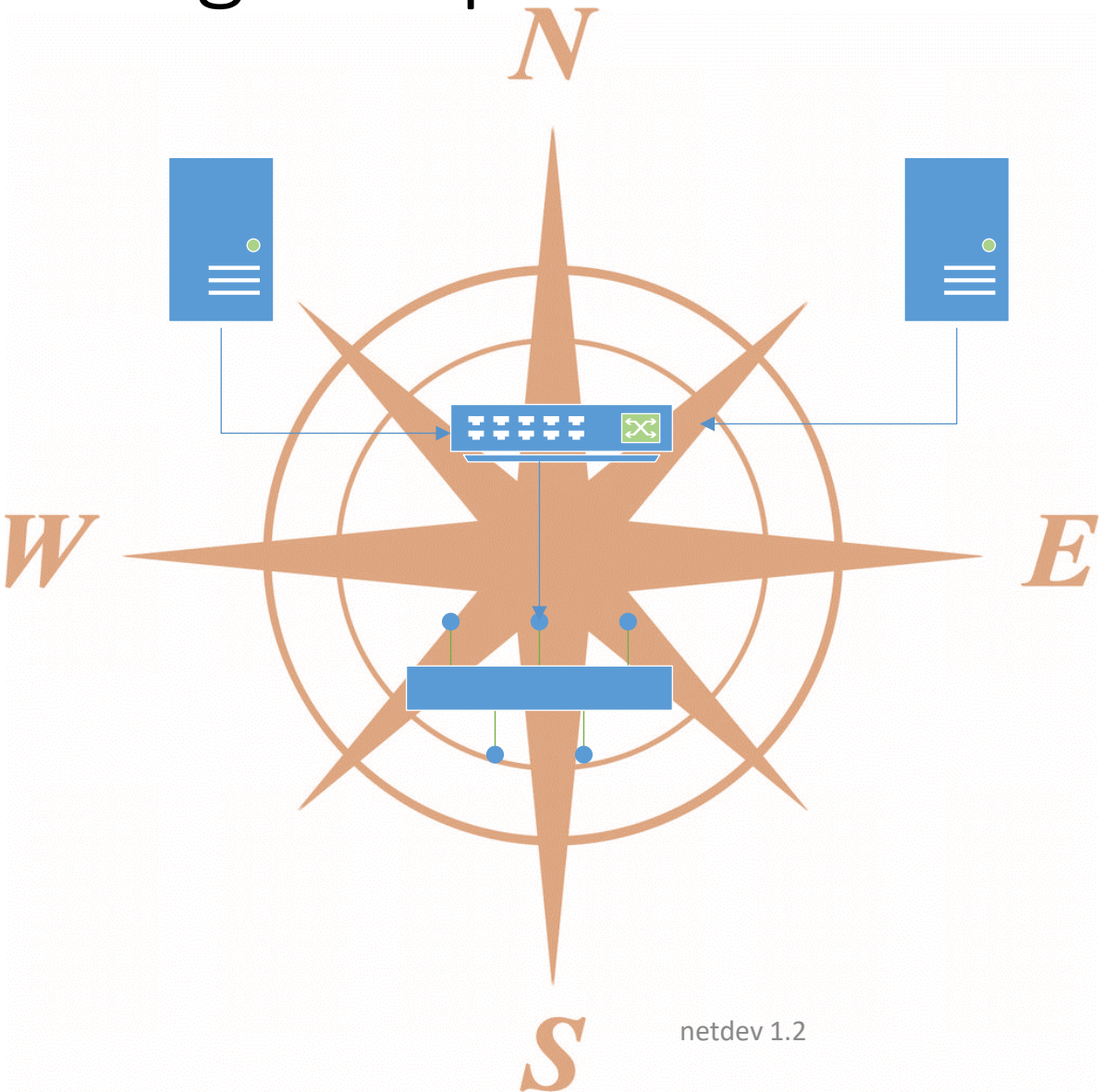

PRETTY, ISN'T IT?
WHAT IS IT?


I'VE GOT A BUNCH OF VIRTUAL WINDOWS MACHINES NETWORKED TOGETHER, HOOKED UP TO AN INCOMING PIPE FROM THE NET. THEY EXECUTE EMAIL ATTACHMENTS, SHARE FILES, AND HAVE NO SECURITY PATCHES.
BETWEEN THEM THEY HAVE PRACTICALLY EVERY VIRUS..

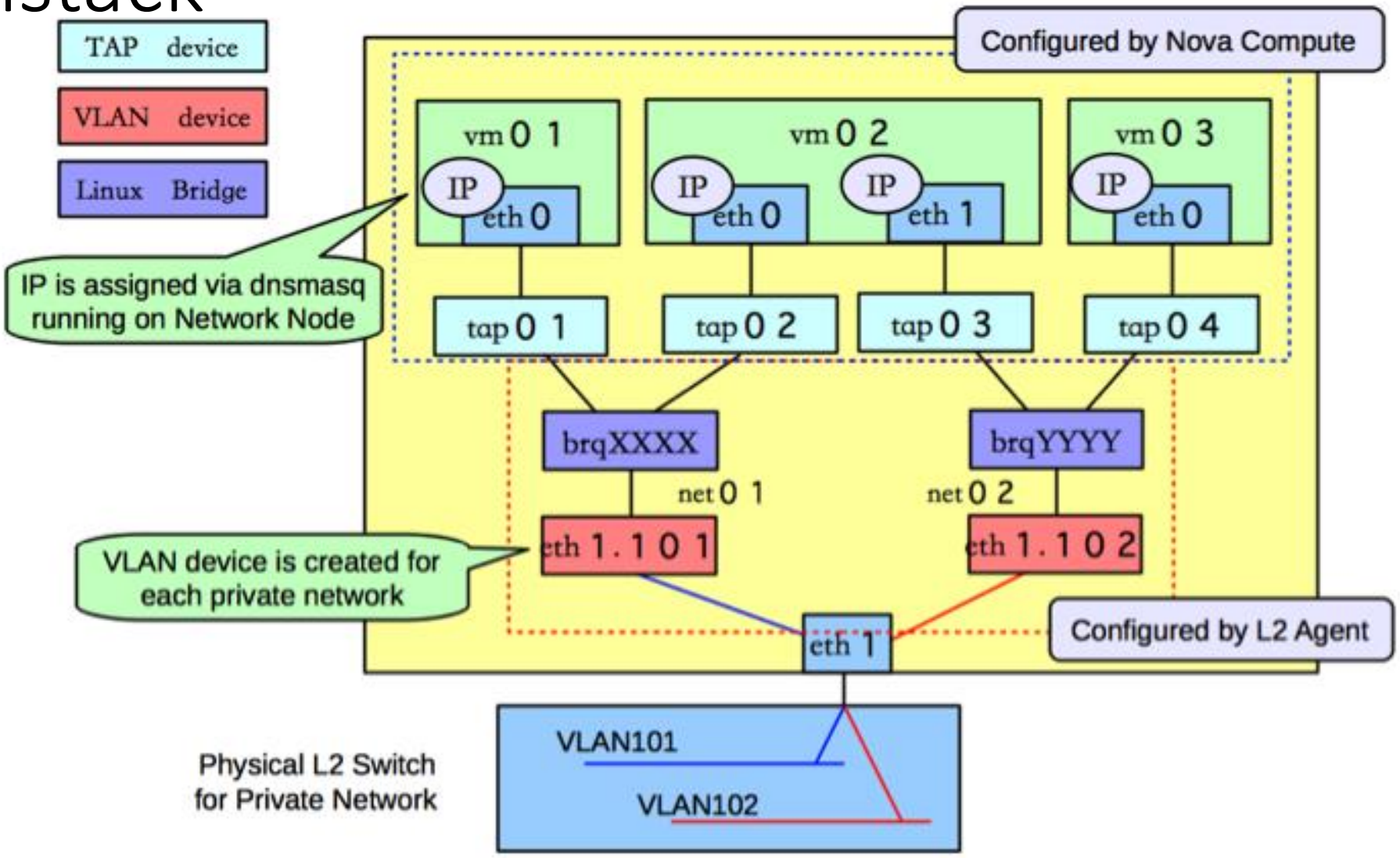

THERE ARE MAIL TROJANS, WARHOL WORMS, AND ALL SORTS OF EXOTIC POLYMORPHICS. A MONITORING SYSTEM ADDS AND WIPES MACHINES AT RANDOM. THE DISPLAY SHOWS THE VIRUSES AS THEY MOVE THROUGH THE NETWORK.
GROWING AND STRUGGLING.


YOU KNOW, NORMAL PEOPLE JUST HAVE AQUARIUMS.
GOOD MORNING, BLASTER. ARE YOU AND W32.WELCHIA GETTING ALONG?
WHO'S A GOOD VIRUS? YOU ARE! YES. YOU ARE!

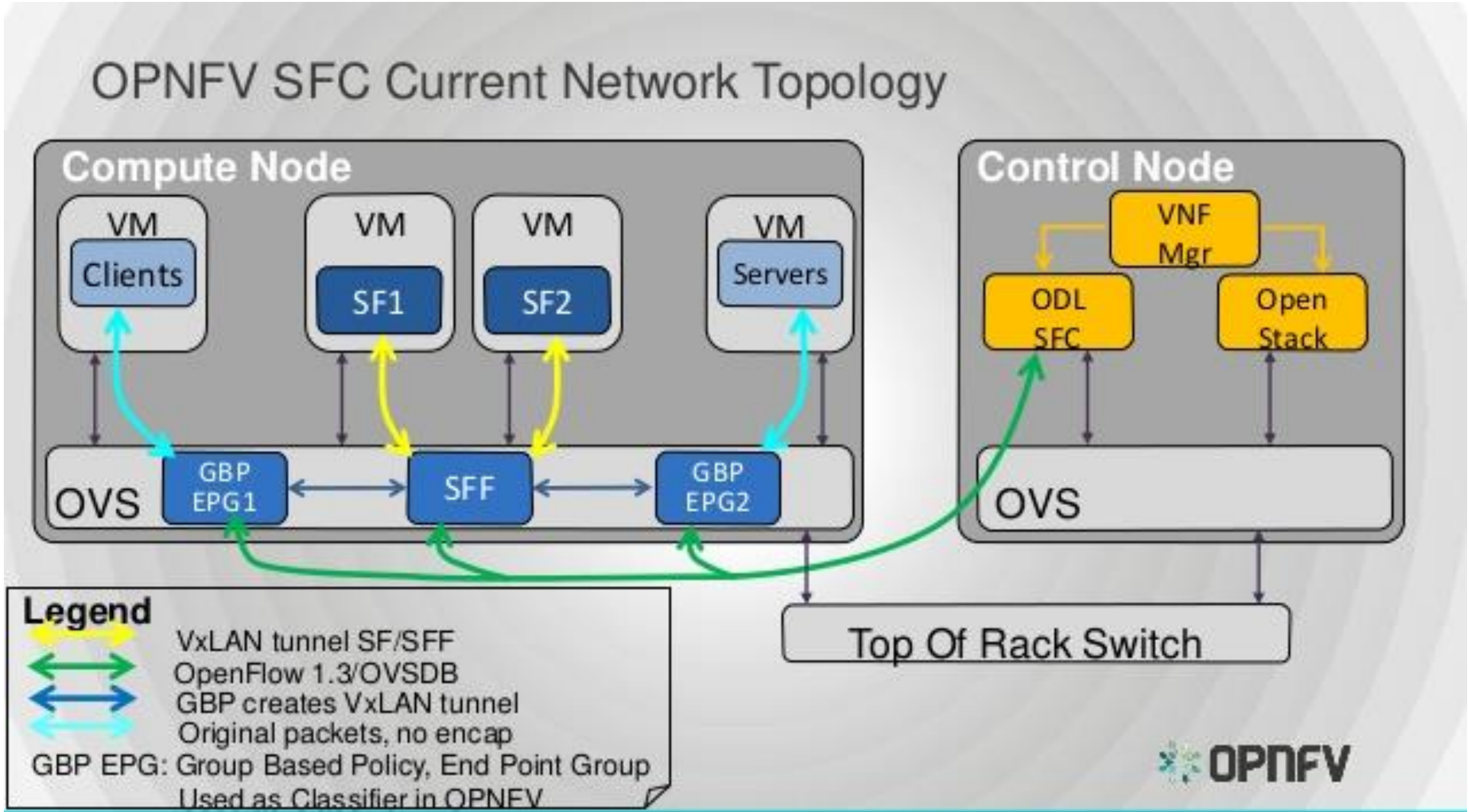
Networking compass directions



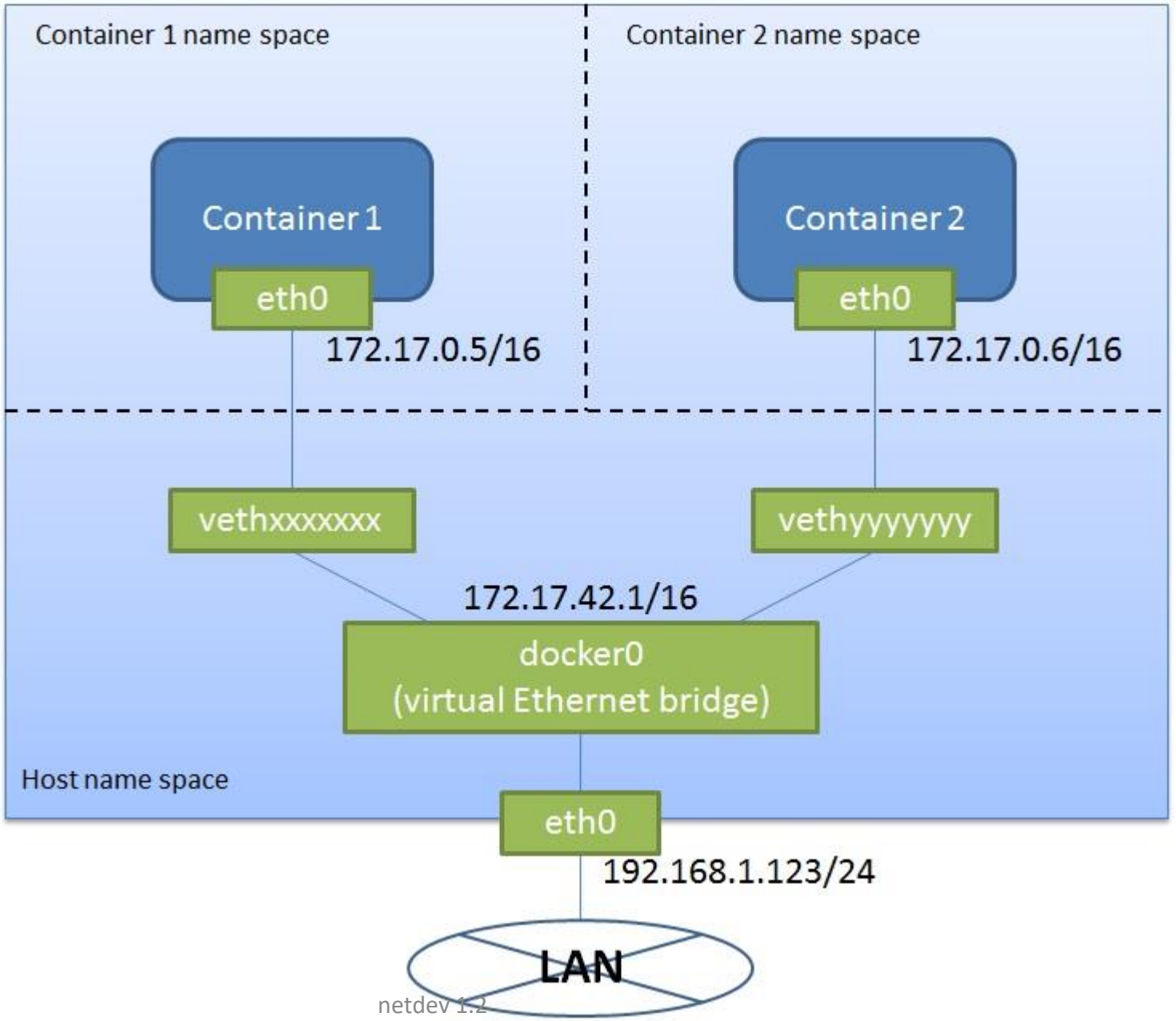
Openstack


TAP device
VLAN device
Linux Bridge
Configured by Nova Compute
vm 0 1
vm 0 2
vm 0 3
IP
eth 0
IP
eth 0
IP
eth 1
IP
eth 0
IP is assigned via dnsmasq running on Network Node
tap 0 1
tap 0 2
tap 0 3
tap 0 4
brqXXXX
brqYYYY
net 0 1
net 0 2
VLAN device is created for each private network
eth 1.1 0 1
eth 1.1 0 2
eth 1
Configured by L2 Agent
Physical L2 Switch for Private Network
VLAN101
VLAN102

OPNFV SFC Current Network Topology



Docker

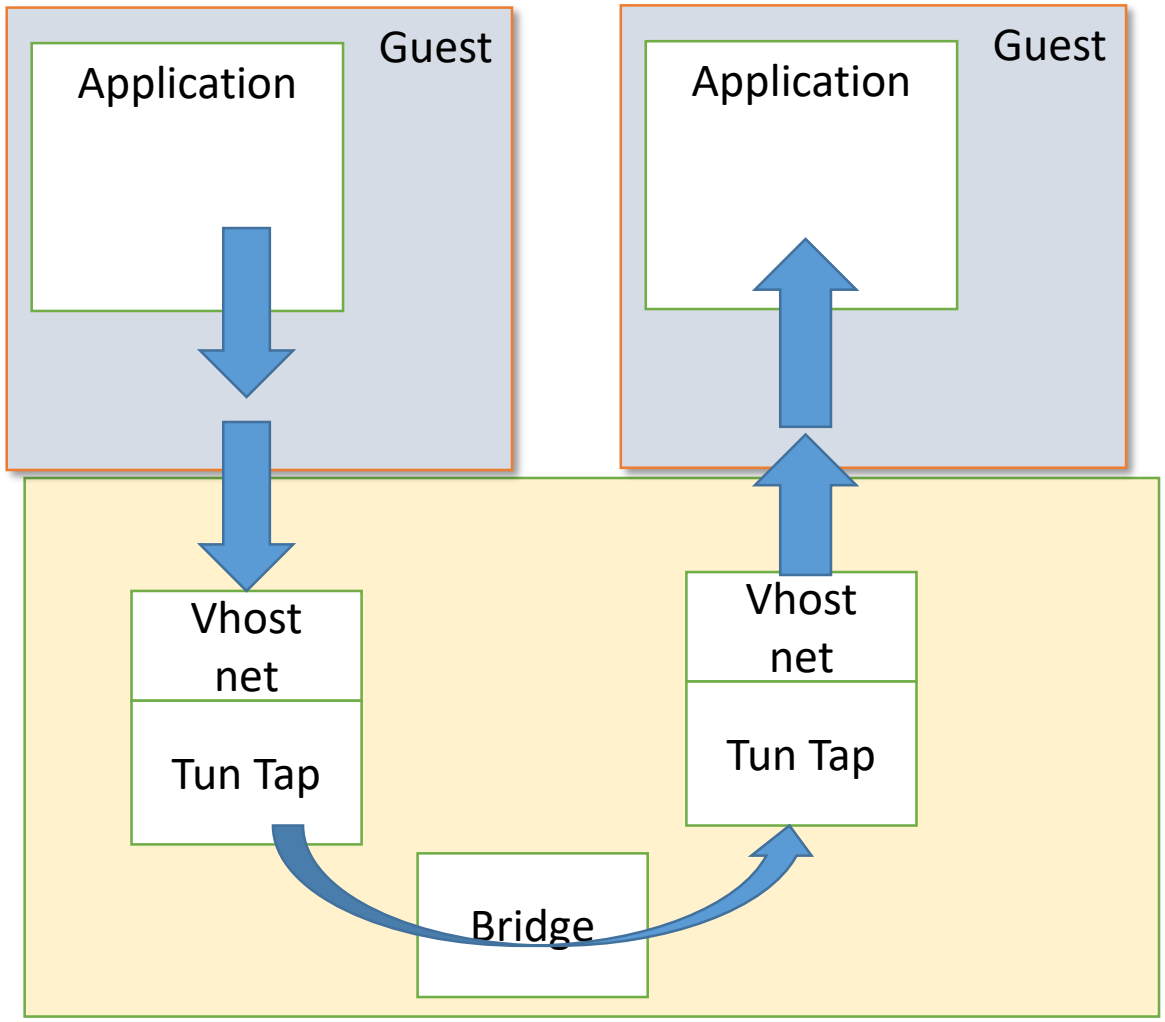

Container 1 name space
Container 2 name space
Container 1
eth0
172.17.0.5/16
Container 2
eth0
172.17.0.6/16
vethxxxxxxx
vethyyyyyyy
172.17.42.1/16
docker0
(virtual Ethernet bridge)
Host name space
eth0
192.168.1.123/24
LAN

	Intel ixgbe	Xen netfront	Vmware vmxnet3	KVM virtio	Hyper-V netvsc
Max MTU	9728	65535	9000	65535	65536
Rx queues	128	any	8	any	any
Tx queues	128	any	8	any	any
LRO	No	No	Yes	No	No
GSO	Yes	Yes	Yes	Yes	Yes
BQL	Yes	No	No	No	No
NAPI	Yes	Yes	Yes	Yes	No
GRO	Yes	Yes	Yes	Yes	No
Multicast filter	Yes	No	Yes	Yes	No
VLAN filter	Yes	No	Yes	Yes	No
Rx Checksum	Yes	Yes	Yes	Yes	Yes
Tx Checksum	Yes	Yes	Yes	Yes	Yes

System Configuration

- CPU Intel Xeon E5-1620 v4 Broadwell-EP 3.5 Ghz
- Memory DDR x 4 bank = 52000 MByte/s
- Host KVM with 4.4 kernel (Ubuntu LTS)
- Guest Ubuntu and Debian guest

Single stream East/West KVM



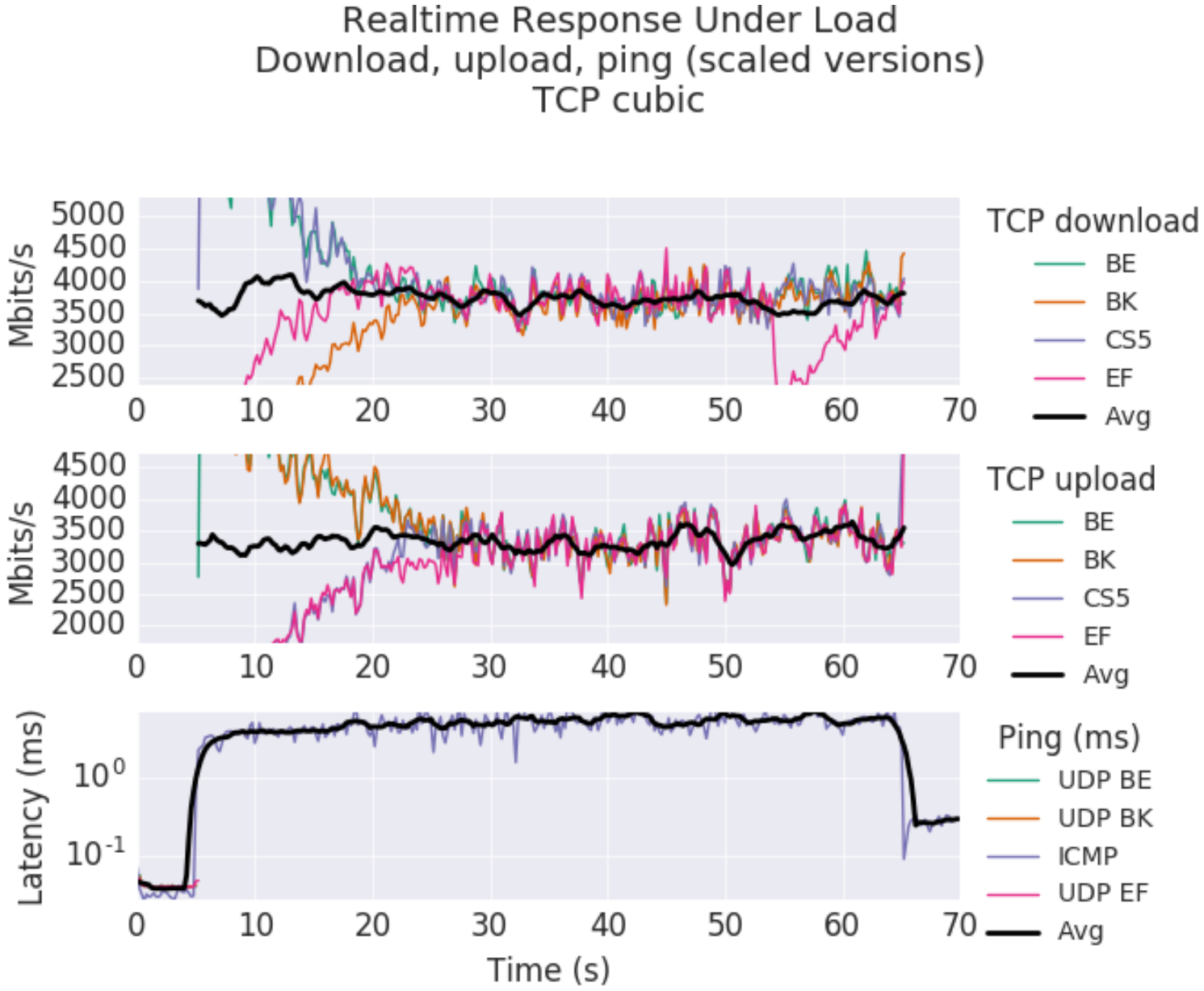
Single stream Bandwidth

- Theoretical
  - 4 copies, 2 syscall and 2 VM exit per RTT
  - Packet size / memcpy rate = 64K bytes/ 52000 MB/s = 1.2us
  - VM exit = .5us
  - 147K TSO packets/sec = 9.6 GB/sec = 77 Gbit/sec
- Measured
  - Iperf single stream => 41 Gbit/sec

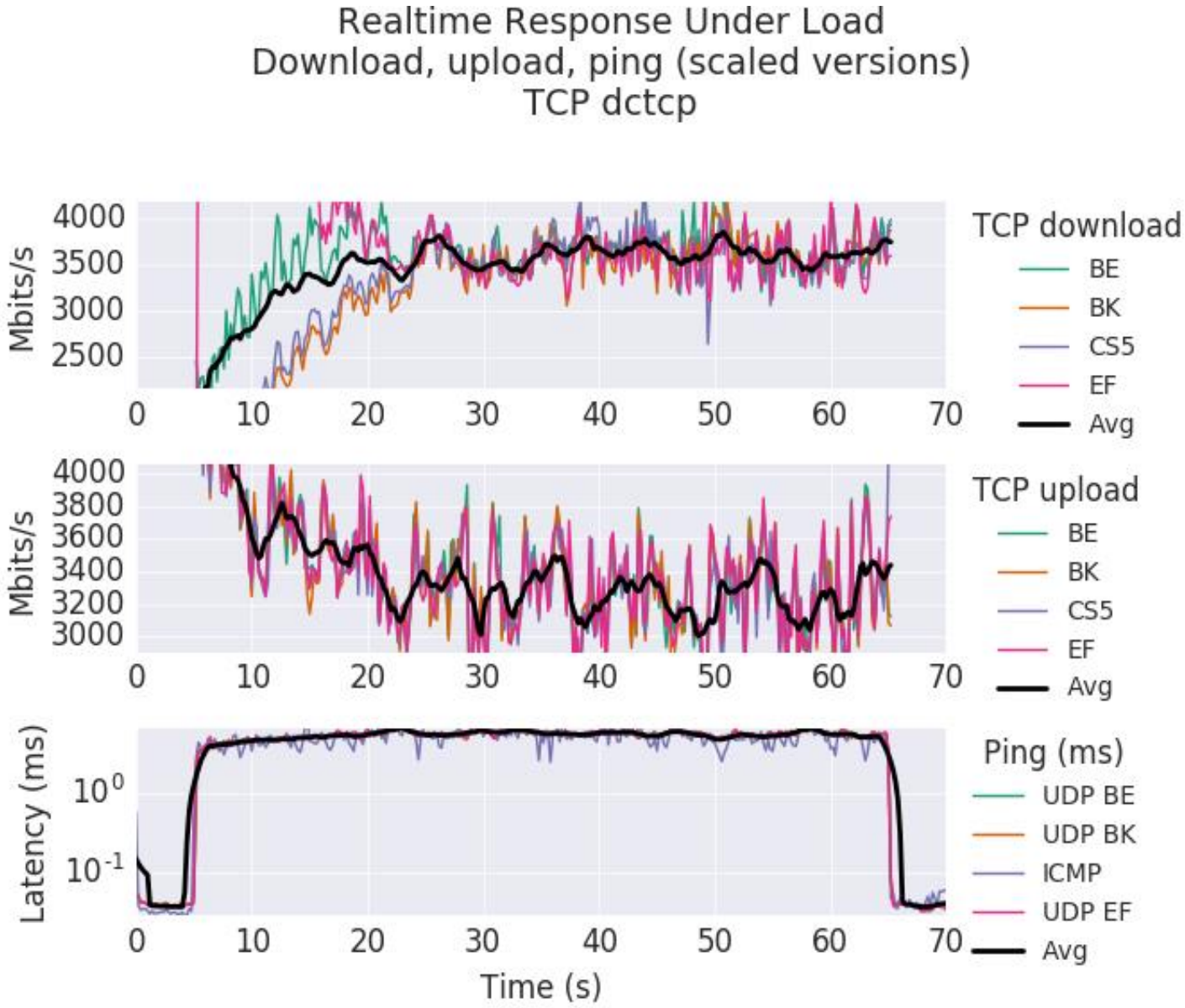
Offloads

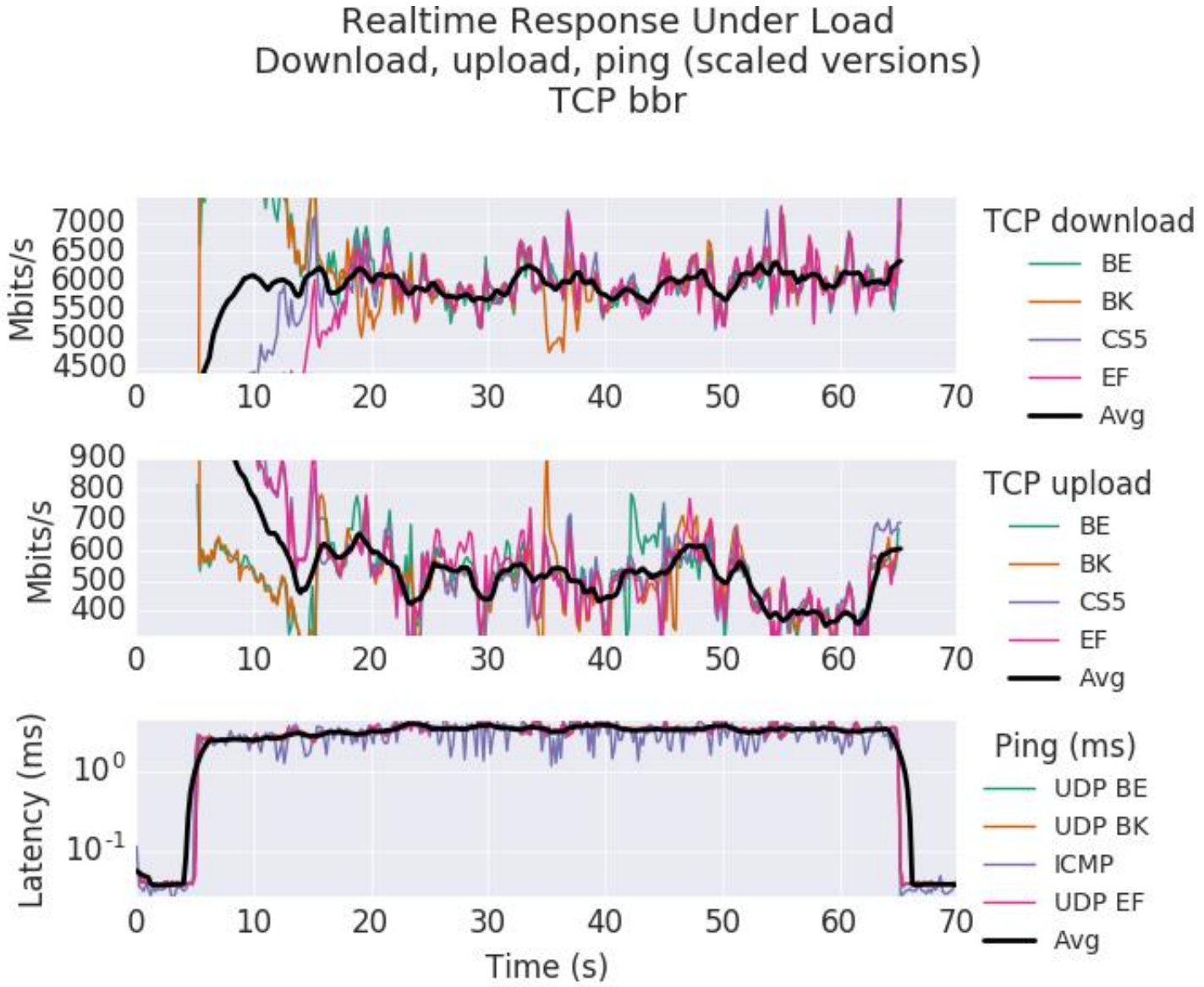
- TCP segmentation offload – no change
- GRO – no change
- Transmit checksum
  - Disabled => 7 Gbit/sec
- SR-IOV => 17 /Gbit/sec

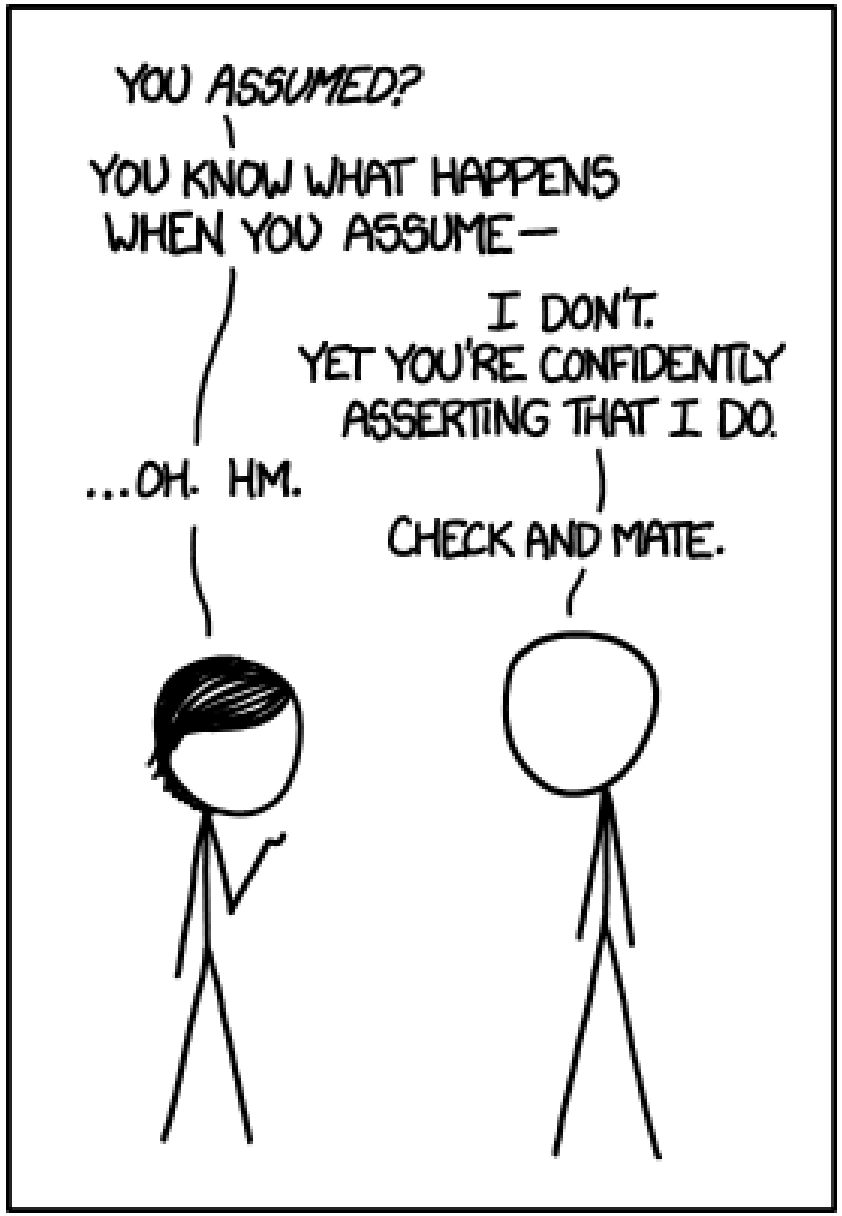
- Tx nocache copy
  - Enabled => 34.4 Gbit/sec
- Rx busy poll
  - Enabled => Guest crash (lockup)



Local/remote: ubuntu-guest1/192.168.122.250 - Time: 2016-09-23T15:15:46.967712 - Length/step: 60s/0.20s


Realtime Response Under Load
Download, upload, ping (scaled versions)
TCP dctcp
Mbits/s
4000
3500
3000
2500
0
10
20
30
40
50
60
70
TCP download
BE
BK
CS5
EF
Avg
Mbits/s
4000
3800
3600
3400
3200
3000
TCP upload
Latency (ms)
10^0
10^{-1}
Ping (ms)
UDP BE
UDP BK
ICMP
UDP EF
Time (s)
Local/remote: ubuntu-guest1/192.168.122.250 - Time: 2016-09-23T15:32:27.225132 - Length/step: 60s/0.20s


Realtime Response Under Load
Download, upload, ping (scaled versions)
TCP bbr
Mbits/s
7000
6500
6000
5500
5000
4500
0
10
20
30
40
50
60
70
TCP download
BE
BK
CS5
EF
Avg
Mbits/s
900
800
700
600
500
400
TCP upload
Latency (ms)
10^0
10^{-1}
Ping (ms)
UDP BE
UDP BK
ICMP
UDP EF
Time (s)
Local/remote: ubuntu-guest1/192.168.122.250 - Time: 2016-09-23T15:16:58.782375 - Length/step: 60s/0.20s


YOU ASSUMED?
YOU KNOW WHAT HAPPENS WHEN YOU ASSUME—
I DON'T. YET YOU'RE CONFIDENTLY ASSERTING THAT I DO.
...OH. HM.
CHECK AND MATE.

Do we have to standard TCP/IP?

- MTU = 1500?
  - What about seeding PMTU?
- Congestion control
  - Does existing IW and CWND make sense here?
- Custom socket family
  - AF_VM, AV_VSOCK, AF_HV

Linux enhancements?

- Virtio 1.1
  - Batching
  - Byte Queue Limits?
- Bulk receive?

Do we really need kernel/user copy?

- Direct I/O
  - Splice + mmap hack
  - AIO
- Do networking in userspace
  - DPDK – separate drivers, thread model, …
  - TLDK – TCP/IP in userspace
- Eliminate kernel
  - Unikernel – Cloudius, RUMP kernel, …

Is vswitch necessary?

- Shared memory
  - Security
  - What about encrypted VM
- Vswitch in hardware?

Thank You

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